

DIGITAL TELEVISION DEMYSTIFIED

India is readying for a complete shift from analogue to digital TV transmission. Digital television will provide better picture and sound quality and greater programme choice. Let's take a look at what goes into digital TV broadcasting



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Broadcasting a high volume of information requires a very wide bandwidth, which for analogue television is five to six megahertz. For digital video broadcasting (DVB) ten times the analogue bandwidth or more is necessary. For this reason, data compression techniques are used to reduce the bandwidth so that more than one channel can be transmitted in the space allocated to a single channel for analogue TV.

There are many advantages of digi-

tal TV over analogue, such as:

1. Good picture quality
2. Increased number of channels within the same bandwidth
3. Lower transmission power
4. Reduced adjacent-channel interference

Broadcasting of digital TV signals involves digitisation, compression and channel encoding. Let's go through digitisation and compression mechanisms in detail.

Digitisation

Digitisation is the process of converting the analogue audio and video

signals into a series of bits using an analogue-to-digital converter (ADC). Digitising a TV picture means sampling the contents of the picture frame-by-frame and scan-line-by-scan-line. In order to maintain the quality of the picture, there must be at least as many samples per line as there are pixels, with each sample representing one pixel.

Sampling of TV signal

Sampling refers to the process of converting a continuous analogue signal into discrete digital numbers. Typically, an ADC is used to convert voltages into digital numbers. The process may be reversed through a digital-to-analogue converter (DAC).

Sampling rate. In order to convert analogue signals (which are continuous in time) into digital representation, these must be sampled at discrete intervals in time. The interval at which the signal is captured is known as the sampling rate of the converter.

If the sampling rate is fast enough, the stored sampled data points may be used to reconstruct the original signal exactly from the discrete data by interpolating the data points. Ultimately, the accuracy of the reconstructed signal is limited by the quantisation error, and is possible only if the sampling rate is higher than twice the highest frequency of the signal. This is the basis for the 'Shannon-Nyquist Sampling Theorem.' If the signal is not sampled at baseband, it must be sampled at greater than twice